

ORIGINAL RESEARCH ARTICLE

Heart Rate and Cardiovascular Outcomes in Post-Myocardial Infarction Patients Treated by β -Blockers: A Secondary Analysis of the ABYSS Trial

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BACKGROUND: Heart rate (HR) is a key prognostic factor after myocardial infarction (MI), but its relevance in the modern reperfusion era is uncertain. We aim to evaluate the association between HR and β -blocker interruption on cardiovascular outcomes.

METHODS: A prespecified secondary analysis of the ABYSS trial (Assessment of Beta-Blocker Interruption 1 Year After an Uncomplicated Myocardial Infarction), including 3698 stable post-MI patients (left ventricular ejection fraction $\geq 40\%$) randomized to continue or interrupt β -blockers, was conducted. Patients were grouped by prerandomization HR tertiles: <60 bpm (T1), 60 to <68 (T2), and ≥ 68 (T3). We examined associations between HR, treatment strategy, and the primary endpoint (death, MI, stroke, or cardiovascular rehospitalization), major secondary endpoints, and on-treatment HR.

RESULTS: Median age in the study population was 63.5 years (55.9–71.1), and there were 621 women (17.1%). Baseline HR was not associated with the primary endpoint (22.4% versus 21.8% versus 21.6%; $P=0.867$). Higher HR was associated with increased risk of death, MI, or stroke (5.5% versus 6.4% versus 9.2%; $P<0.001$; T3 versus T1 adjusted hazard ratio, 1.55; 95% CI, 1.14–2.12) and death, MI, stroke, or heart failure (6.5% versus 7.1% versus 10.4%; $P=0.007$; T3 versus T1 adjusted hazard ratio, 1.47; 95% CI, 1.11–1.97). All-cause mortality rose across tertiles (2.9% versus 3.4% versus 5.9%; $P=0.004$; P trend=0.008). β -Blocker interruption produced a dose-dependent HR increase of ≈ 10 –13 bpm during follow-up. The association between interruption and worse outcomes was consistent across HR tertiles (no significant interaction) and LVEF categories (40% to 49% and $>50\%$).

CONCLUSIONS: In stabilized post-MI patients with preserved ejection fraction, higher HR remains associated with adverse cardiovascular events and mortality in the reperfusion era. Interrupting β -blockers substantially increases HR and is consistently linked with worse outcomes irrespective of baseline HR, supporting continuation of β -blocker therapy.

Key Words: β -blockers ■ heart rate ■ myocardial infarction ■ outcomes

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCULATIONAHA.125.078635>.

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Clinical Perspective

What Is New?

- In this prespecified secondary analysis of the ABYSS randomized trial, higher baseline heart rate under β -blocker therapy was associated with increased risk of major cardiovascular events and mortality in stable post-myocardial infarction (MI) patients with left ventricular ejection fraction $\geq 40\%$.
- β -Blocker interruption resulted in a sustained increase in heart rate of ≈ 10 – 13 bpm during follow-up.
- The association between β -blocker interruption and worse clinical outcomes was consistent across baseline heart rate tertiles, with no significant interaction.

What Are the Clinical Implications?

- These findings support the prognostic relevance of resting heart rate in stable post-MI patients in the contemporary reperfusion era.
- The consistent increase in heart rate after β -blocker discontinuation suggests that heart rate control may contribute to the protective effects of β -blockers after myocardial infarction.
- In stable post-MI patients with preserved or mildly reduced left ventricular ejection fraction, discontinuation of well-tolerated β -blocker therapy should be approached cautiously.

Hear rate (HR) has long been a therapeutic target in post-myocardial infarction (MI) management, dating back to the advent of modern coronary care units.^{1,2} Elevated HR after MI is associated with increased myocardial oxygen consumption, adverse ventricular remodeling, and autonomic dysregulation, all of which contribute to higher morbidity and mortality.^{3–5} Consequently, β -blockers have become a cornerstone of post-MI care, aiming to reduce HR and myocardial ischemia and thereby improve outcomes.^{6,7}

The widespread adoption of early revascularization has profoundly changed MI management, leading to smaller infarct sizes, fewer arrhythmic complications, and reduced heart failure (HF) incidence. As a result, the long-term use of β -blockers in stable post-MI patients, especially those with preserved left ventricular ejection fraction (LVEF), has become a subject of debate, particularly given concerns about quality of life (QoL) and safety.^{8,9}

Recently, the ABYSS trial demonstrated that discontinuing β -blockers several months or years after MI in stable patients with an LVEF $\geq 40\%$ was associated with an increased risk of cardiovascular events and no improvement in QoL, supporting their continued use in this population.¹⁰ Trials evaluating the early initiation of β -blockers in intensive care unit patients with a post-MI

Nonstandard Abbreviations and Acronyms

ABYSS	Assessment of Beta-Blocker Interruption 1 Year after an Uncomplicated Myocardial Infarction on Safety and Symptomatic Cardiac Events Requiring Hospitalization
ACE	angiotensin-converting enzyme
ARB	angiotensin receptor blocker
CAD	coronary artery disease
CV	cardiovascular
EQ-5D	EuroQol 5-Dimension questionnaire
HF	heart failure
HR	heart rate
IQR	interquartile range
LVEF	left ventricular ejection fraction
MACE	major adverse cardiovascular events
MI	myocardial infarction
NSTEMI	non-ST-elevation myocardial infarction
PHRC	Programme Hospitalier de Recherche Clinique
QoL	quality of life
RCS	restricted cubic splines
SD	standard deviation
STEMI	ST-elevation myocardial infarction
T1	tertile 1
T2	tertile 2
T3	tertile 3
TIA	transient ischemic attack

LVEF $>40\%$ have yielded contrasting results: whereas the BETAMI-DANBLOCK trial demonstrated a benefit, the more recent REDUCE-AMI trial (Randomized Evaluation of Decreased Usage of Beta-Blockers after Acute Myocardial Infarction) and the REBOOT-CNIC trial showed no significant advantage.^{11–13}

Following this ongoing debate, this prespecified analysis of the ABYSS trial aims to further investigate the mechanisms underlying the increased cardiovascular risk associated with β -blocker discontinuation, specifically by examining the effect of baseline HR under β -blocker therapy on cardiovascular outcomes and the effect of β -blocker discontinuation over time on HR and cardiovascular events.

METHODS

Study Design and Participants

The data that support the findings of this study are available from the corresponding author upon reasonable request. This was a prespecified secondary analysis of the ABYSS trial (NCT03498066 et EudraCT 2017-003903-23). The study design and main results of the ABYSS trial have been described

previously.¹⁴ Briefly, ABYSS was a multicenter, randomized, open-label, noninferiority trial with blinded adjudication of all endpoints (PROBE [Prospective Randomized Open, Blinded End-Point trial] study design), conducted at 49 clinical sites in France. Participants were considered eligible if they had experienced an MI at least 6 months earlier, were receiving β -blocker therapy, had a preserved LVEF $>40\%$, and no clinical HF. The main exclusion criteria were chronic HF or reduced LVEF ($\leq 40\%$), any cardiac event in the past 6 months, and any other primary indication for β -blocker therapy (arrhythmia, migraine, HF, uncontrolled hypertension, persistent angina, or ischemia $>10\%$). Individuals who accepted to participate in the trial were randomized to β -blocker therapy interruption (experimental arm) versus β -blocker therapy continuation (control arm). The CONSORT (Consolidated Standards of Reporting) 2025 checklist is available at the end of the [Supplemental Material](#). There was no patient or public involvement in the design, conduct, and reporting of the trial. The patients underwent randomization between August 28, 2018, and September 12, 2022.

Objectives and Analyses

The first set of analyses examined the effect of baseline HR (HR at the moment of randomization) on the primary and secondary endpoints of the ABYSS trial. The ABYSS primary endpoint was a composite of all-cause death, nonfatal MI, nonfatal stroke, or hospitalization for any cardiovascular reason, assessed in a time-to-first event. Main secondary endpoints were: (1) all-cause death, nonfatal MI, or nonfatal stroke; (2) all-cause death, nonfatal MI, nonfatal stroke, or acute HF; and (3) each endpoint taken separately.

The second set of analyses assessed the interaction between baseline HR and effect of β -blocker interruption on the study outcomes (primary and secondary endpoints of the ABYSS trial).

The third set of analyses examined the effect of β -blocker interruption on HR along the follow-up, stratified by baseline HR and initial β -blocker dose. All patients were followed at 6 and 12 months, then annually until the last patient had completed at least 1 year of follow-up. Resting HR was measured at each visit.

The protocol was approved by national and institutional regulatory and ethical authorities, and all patients provided written informed consent.

Statistical Analysis

All analyses were performed according to the intention-to-treat principle. Categorical variables are presented as counts (percentages), and continuous variables as means with standard deviations (SDs) or medians with interquartile ranges (IQRs, 25th–75th percentiles), as appropriate. Baseline characteristics were compared across baseline HR tertiles using Pearson's χ^2 or Fisher's exact test for categorical variables, and analysis of variance or Kruskal–Wallis tests for continuous variables, as appropriate.

Kaplan–Meier curves were generated to evaluate the association between baseline HR and the primary and secondary endpoints of ABYSS. Differences between HR tertiles were assessed using log-rank tests. The association between β -blocker interruption and study endpoints was assessed using Kaplan–Meier curves stratified by baseline HR tertiles and randomization group, with censoring at the date of last follow-up.

Univariable and multivariable Cox proportional hazards models were used to assess factors associated with the primary endpoint. Baseline characteristics associated with the outcome at a significance level of $P < 0.20$ in univariable analyses were considered for inclusion in the multivariable Cox proportional hazards model. Hazard ratios with 95% CIs were estimated from Cox proportional hazards models adjusted for age, current smoking, diabetes mellitus, non–ST-elevation myocardial infarction (NSTEMI), multivessel disease, peripheral artery disease, previous stroke or transient ischemic attack, previous HF, left ventricular ejection fraction, and residual angina.

To assess trends across baseline HR tertiles, P trend values were derived from the adjusted Cox proportional hazards models by entering HR tertiles as an ordinal variable.

To assess whether the effect of β -blocker interruption differed according to baseline HR categories, interaction analyses were performed including an interaction term between treatment group and baseline HR tertiles. The reported P value corresponds to the global test of interaction across the 3 HR categories.

The proportional hazards assumption was assessed using scaled Schoenfeld residuals and graphical methods. Individual outcomes were analyzed using Fine–Gray sub distribution hazard models, treating death as a competing risk when appropriate.

To assess the robustness of the main results, sensitivity analyses were performed modeling baseline HR as a continuous variable using restricted cubic splines (RCS). Adjusted Cox proportional hazards models with RCS terms were used to flexibly examine the association between baseline HR and clinical outcomes. Knots were placed according to Harrell's recommended percentiles, and the median baseline HR (63 bpm) was used as the reference value. The optimal number of knots was selected based on the Akaike information criterion. Restricted cubic spline models were also used to explore potential interaction effects.

Additional stratified analyses were conducted according to left ventricular ejection fraction (40% to 49% and $\geq 50\%$).

Changes in HR during follow-up were also analyzed according to β -blocker dose categories ($<25\%$, 25% to 49%, 50% to 74%, and $\geq 75\%$ of target dose) and treatment group. In this secondary analysis, missing data were not imputed. Patients lost to follow-up were censored at the date of last contact.

All hypothesis tests were 2-sided, and a P value < 0.05 was considered statistically significant. Statistical analyses were performed using SAS software, version 9.4 (SAS Institute).

RESULTS

Baseline Characteristics

Among 3623 patients with available HR data (98% of the cohort), 1220 had HR < 60 bpm (median 55), 1149 had HR 60–67 bpm (median 63), and 1254 had HR ≥ 68 bpm (median 75). Baseline characteristics stratified by HR tertiles are presented in Table 1. Patients in the highest HR tertile (T3) were younger (median age 64.3 years in T1, 63.1 years in T2, and 61.8 years in T3; $P < 0.001$), more frequently female (13.6%, 16.7%, 20.2%; $P < 0.001$) more often smoker (15.2%, 20.3%,

Table 1. Baseline Characteristics*

Median (IQR) or No. (%)	T1 <60 bpm N=1220	T2 60 to <68 bpm N=1149	T3 ≥68 bpm N=1254	P value
Age, y	64.3 (57.8; 71.2)	64.2 (56.5; 72.0)	61.8 (54.0; 70.0)	<0.001
Male	1054 (86.4%)	947 (82.4%)	1001 (79.8%)	<0.001
Body mass index, kg/m ²	26.0 (23.8; 28.4)	26.3 (24.0; 29.1)	27.3 (24.2; 30.8)	<0.001
Current smoker	186 (15.2%)	208 (18.1%)	324 (25.8%)	<0.001
Diabetes mellitus	143 (11.7%)	222 (19.3%)	371 (29.6%)	<0.001
Dyslipidemia	626 (51.3%)	628 (54.7%)	648 (51.7%)	0.204
Family history of CAD	367 (30.1%)	308 (26.8%)	369 (29.4%)	0.179
Renal insufficiency	59/1052 (5.6%)	47/987 (4.8%)	110/1087 (10.1%)	<0.001
Past medical history				
STEMI	787 (64.5%)	731 (63.6%)	769 (61.3%)	0.238
Duration between last MI and randomization, y	2.7 (1.1; 6.0)	2.9 (1.2; 7.1)	3.0 (1.3; 6.3)	0.053
Multivessel disease	638 (52.3%)	607 (52.8%)	644 (51.4%)	0.764
Revascularization for the MI events	1158 (94.9%)	1099 (95.6%)	1188 (94.7%)	0.555
Peripheral artery disease	65 (5.3%)	44 (3.8%)	75 (6.0%)	0.05
Stroke or TIA	35 (2.9%)	35 (3.0%)	51 (4.1%)	0.202
Previous episode of heart failure	12 (1.0%)	20 (1.7%)	28 (2.2%)	0.05
Health status at baseline				
Median (IQR) systolic at baseline, mm Hg	130 (120; 142)	131 (121; 143)	134 (123; 146)	<0.001
Median (IQR) diastolic at baseline, mm Hg	75 (70; 80)	77 (70; 82)	80 (72; 86)	<0.001
Median (IQR) HR at baseline, bpm	55 (52; 57)	63 (61; 65)	75 (71; 81)	<0.001
Median (IQR) LVEF (%)	60 (51; 60)	58 (52; 60)	60 (53; 60)	0.249
LVEF 40–49%	118 (9.6%)	105 (9.3%)	119 (10.0%)	
LVEF ≥50%	1115 (90.4%)	1019 (90.7%)	1070 (90.0%)	
Residual angina	15 (1.2%)	14 (1.2%)	20 (1.6%)	0.655
HR at baseline, bpm	55 (52; 57)	63 (61; 65)	75 (71; 81)	<0.001
Treatment at randomization				
Statin	1164 (95.4%)	1103 (96.0%)	1193 (95.1%)	0.586
Ezetemibe	356 (29.2%)	337 (29.3%)	337 (26.9%)	0.319
ACE or ARB	910 (74.6%)	831 (72.3%)	935 (74.6%)	0.357
Calcium blockers	131 (10.7%)	146 (12.7%)	158 (12.6%)	0.245
Diuretics	120 (9.8%)	100 (8.7%)	160 (12.8%)	0.003
Aldacton or eplerenon (%)	36 (3.0%)	27 (2.3%)	35 (2.8%)	0.649

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; bpm, beats per minute; bpm, beats per minute; CAD, coronary artery disease; HR, heart rate; IQR, interquartile range; LVEF, left ventricular ejection fraction; MI, myocardial infarction; STEMI, ST-elevation myocardial infarction; T1/T2/T3, tertile 1/tertile 2/tertile 3; and TIA, transient ischemic attack.

and 25.8%; $P<0.001$) had higher prevalence of diabetes (11.7%, 18.4%, and 29.6%; $P<0.001$) and renal insufficiency (5.6% versus 7.9% versus 10.1%; $P<0.001$). β -Blocker distribution at randomization and achieved doses are described in Tables S1 and S2. Bisoprolol was the most frequently prescribed β -blocker with comparable distribution across tertiles (73.9% in T1, 71.1% in T2, and 70.4% in T3). Acebutolol and atenolol were likewise distributed across the HR tertiles (acebutolol: 11.3%, 12.4 % and 8.9 %; atenolol 8.5 %, 8.4 % and 9.0 %). Interestingly, patients within the highest tertile of HR (≥ 68 bpm) received higher β -blocker doses with 48.0%

of patients in T3, 43.7% in T2 and 39.2% in T1 receiving $\geq 50\%$ of the target dose.

Cardiovascular Outcomes According to HR Under β -Blocker Therapy

Kaplan Meier estimates for primary and secondary endpoints are displayed in Figure 1. Observed event rates across HR tertiles are presented in Table 2. There was no association between baseline HR and the risk of primary endpoint (22.4 versus 21.8 % versus 21.6%; $P=0.867$). However, higher baseline HR was associated with higher

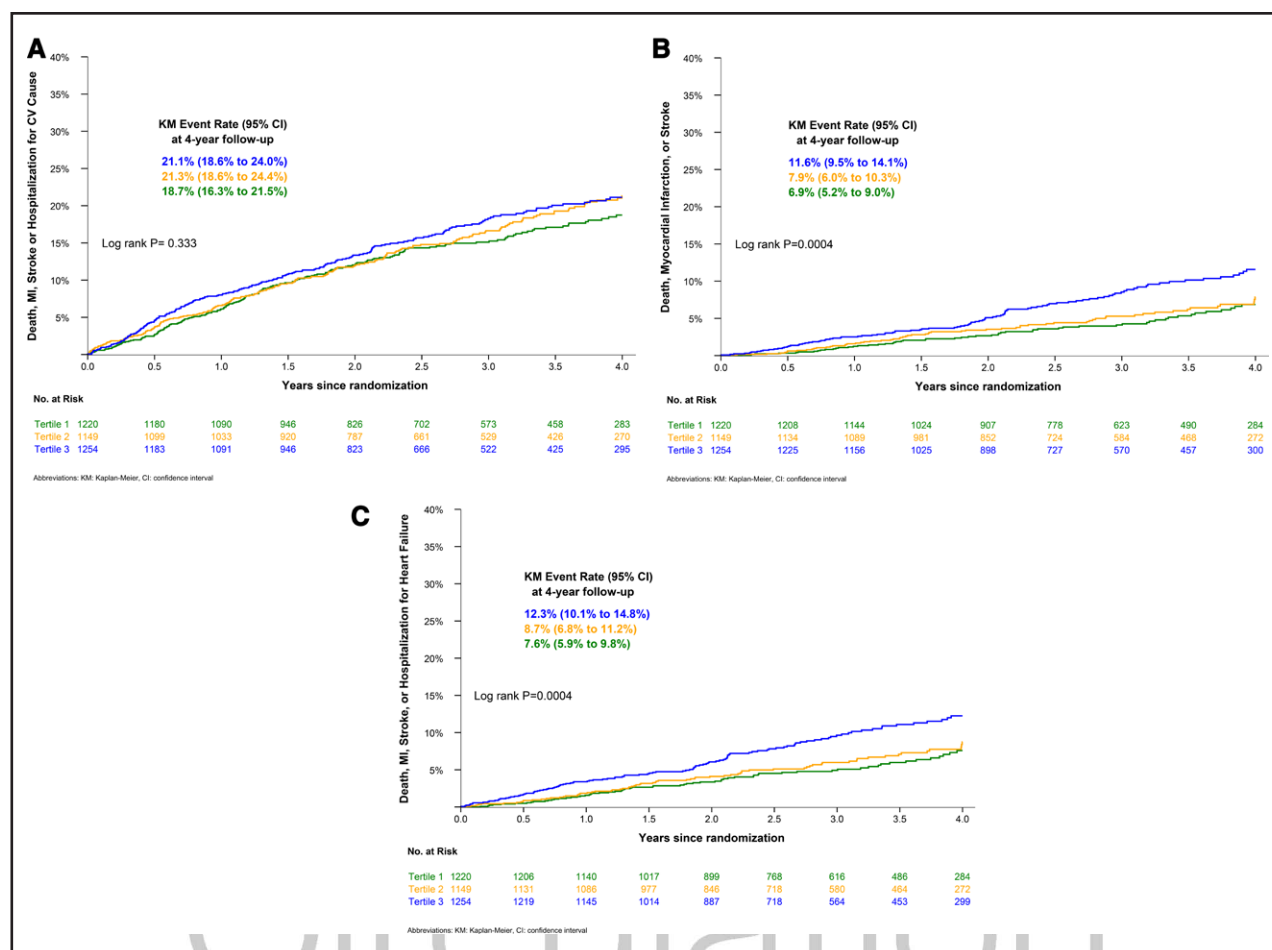


Figure 1. Kaplan-Meier estimates of the cumulative incidence of clinical outcomes according to baseline HR tertiles over a maximum follow-up of ≈4 years.

A, Primary endpoint: death, myocardial infarction, stroke, or cardiovascular hospitalization. **B**, Composite of death, myocardial infarction, or stroke. **C**, Composite of death, myocardial infarction, stroke, or hospitalization for HF. *P* values were derived from log-rank tests comparing the 3 HR tertiles. CV indicates cardiovascular; HF, heart failure; HR, heart rate; KM, Kaplan-Meier; and MI, myocardial infarction.

rates of major adverse cardiovascular events (MACE) defined by the composite of death, MI, or stroke (5.5% versus 6.4% versus 9.2%; $P<0.001$), driven by higher rates of all-cause mortality (2.9% versus 3.4% versus 5.9%; $P=0.004$), cardiovascular death (0.7% versus 1.2% versus 2.1%; $P=0.014$), and nonfatal MI (1.8% versus 2.3% versus 3.3%; $P<0.001$). The composite of MACE or HF was also higher in patient with higher HR (6.5% versus 7.1% versus 10.4%; $P<0.001$). These observations remained consistent after adjustment for confounding variables (Table 2) and when using HR as a continuous variable (Figure S1).

Interaction Between HR and β -Blocker Interruption on Outcomes

The effect of β -blocker interruption on the primary and secondary endpoints was consistent across tertiles of HR with a persistent association with worse outcomes (Table 3). The association between β -blocker and worst

outcomes did not vary in T1, T2, and T3 (P interaction=0.51).

Outcomes According to LVEF

Among the 3546 patients with available LVEF data, 342 (9.6%) had an LVEF 40% to 49%, and 3204 (90.4%) had an LVEF $\geq 50\%$. In the overall population, higher baseline HR was associated with an increased risk of the primary composite endpoint as well as secondary composite endpoints (Figure S1). When analyses were stratified according to LVEF, the association between higher HR and adverse outcomes remained consistent in patients with an LVEF $\geq 50\%$ (Figure S2). In patients with an LVEF 40% to 49%, spline analyses showed similar directional associations, although estimates were less precise because of the smaller sample size (Figure S3). No significant interaction was observed between LVEF category and the association between HR and clinical outcomes.

Table 2. Primary and Secondary Outcomes According to HR at Baseline

Endpoint	T1 <60 bpm N=1220	T2 60 to <68 bpm N=1149	T3 ≥68 bpm N=1254	Adjusted* hazard ratio		P trend
				T3 vs T1 (95% CI, P)	T2 vs T1 (95% CI, P)	
Primary composite (death, MI, stroke, or hospital CV)	263 (21.6%)	250 (21.8%)	281 (22.4%)	0.98 (0.83–1.17), P=0.853	0.95 (0.80–1.13), P=0.569	0.844
Secondary composite (death, MI, or stroke)	67 (5.5%)	73 (6.4%)	115 (9.2%)	1.55 (1.14–2.12), P=0.005	1.07 (0.77–1.50), P=0.677	0.007
Secondary composite (death, MI, stroke, or HF hospitalization)	79 (6.5%)	82 (7.1%)	131 (10.4%)	1.47 (1.11–1.97), P=0.008	1.01 (0.74–1.38), P=0.949	0.007
Death	35 (2.9%)	39 (3.4%)	74 (5.9%)	1.83 (1.21–2.77), P=0.004	1.07 (0.68–1.70), P=0.765	0.004
MI	22 (1.8%)	26 (2.3%)	42 (3.3%)	1.64 (0.98–2.74), P=0.060	1.13 (0.63–2.01), P=0.690	0.114
Stroke	18 (1.5%)	9 (0.8%)	9 (0.7%)	0.45 (0.20–0.99), P=0.047	0.48 (0.21–1.10), P=0.084	0.076
Hospitalization for CV reasons	224 (18.4%)	202 (17.6%)	211 (16.8%)	0.86 (0.71–1.04), P=0.122	0.90 (0.74–1.10), P=0.312	0.29

bpm indicates beats per minute; CV, cardiovascular; HF, heart failure; MI, myocardial infarction; T1/T2/T3, tertile 1/tertile 2/tertile 3. P trend values were obtained from the adjusted Cox proportional hazards model including hazard ratio tertiles as an ordinal variable. Values are No. (%). 95% Wald-type CIs are presented. Fine-Gray proportional hazard model was used with death as competing risk. *Adjusting for age, current smoker, diabetes mellitus, NSTEMI, multivessel disease, peripheral artery disease, stroke or TIA, previous episode of heart failure, ejection fraction, and residual angina.

Effects of β -Blocker Interruption on HR

β -Blocker interruption led to a significant and sustained increase in HR compared with the β -blocker continuation group persisting up to 48 months. At 6 months, the increase was most pronounced in patients with higher baseline HR (T1, +9 bpm; T2, +11 bpm; T3, +11 bpm; Figure 2). When analyzing according to previous β -blocker dose, the HR increase was greatest in patients on higher doses ($\leq 25\%$ target dose +7 bpm; 25% to 50% target dose, +9 bpm; $\geq 50\%$ target dose, +13 bpm at 6 months; Figure 3).

Association Between QoL and HR

At baseline, QoL was lower in patients with the highest HR, with median EQ-5D scores of 0.929 in T1, 0.929

in T2, and 0.910 in T3 ($P < 0.01$). Changes in QoL from baseline to follow-up (peak 6–12 months) were similar across tertiles. No substantial QoL differences were observed between interruption and continuation groups, suggesting minimal impact of treatment strategy on perceived health.

DISCUSSION

The present analysis of the ABYSS trial offers contemporary data on the importance of HR as a prognosis factor and describes in detail the hemodynamic changes after β -blocker interruption and relation to cardiovascular events in stable post-MI patients. Thus, we made the following observations: (1) post-MI patients with elevated HR under β -blocker therapy were more often women

Table 3. Interaction Between Baseline HR and β -Blocker Interruption vs Continuation on the Primary and Secondary Outcomes

Endpoint	Subgroup	% Interruption	% Continuation	Adjusted hazard ratio	P value for interaction
				(95% CI)	
Primary endpoint death, MI, stroke, or CV hospitalization	T1 (<60 bpm)	137/620 (22.1%)	126/600 (21.0%)	1.02 (0.80–1.31)	0.51
	T2 (60 to <68 bpm)	129/557 (23.2%)	121/592 (20.4%)	1.10 (0.85–1.41)	
	T3 (≥ 68 bpm)	153/631 (24.2%)	128/623 (20.5%)	1.25 (0.99–1.58)	
Composite	T1 (<60 bpm)	35/620 (5.6%)	32/600 (5.3%)	0.97 (0.60–1.56)	0.91
	T2 (60 to <68 bpm)	38/557 (6.8%)	35/592 (5.9%)	1.09 (0.69–1.74)	
	T3 (≥ 68 bpm)	58/631 (9.2%)	57/623 (9.1%)	0.97 (0.67–1.41)	
Death, MI, or stroke	T1 (<60 bpm)	45/620 (7.3%)	34/600 (5.7%)	1.21 (0.78–1.90)	0.74
	T2 (60 to <68 bpm)	40/557 (7.2%)	42/592 (7.1%)	0.95 (0.61–1.47)	
	T3 (≥ 68 bpm)	68/631 (10.8%)	63/623 (10.1%)	1.03 (0.73–1.46)	

bpm indicates beats per minute; CV, cardiovascular; HF, heart failure; MI, myocardial infarction; and T1/T2/T3, tertile 1/tertile 2/tertile 3. Values are No. (%) or No./total No. (%). The P value for interaction corresponds to the global test of interaction between treatment group and baseline heart rate tertiles in the Cox proportional hazards model.

Adjustment variables include age, current smoker, diabetes mellitus, non-ST-elevation myocardial infarction (NSTEMI), multivessel disease, peripheral artery disease, stroke or TIA, previous episode of heart failure, ejection fraction, and residual angina.

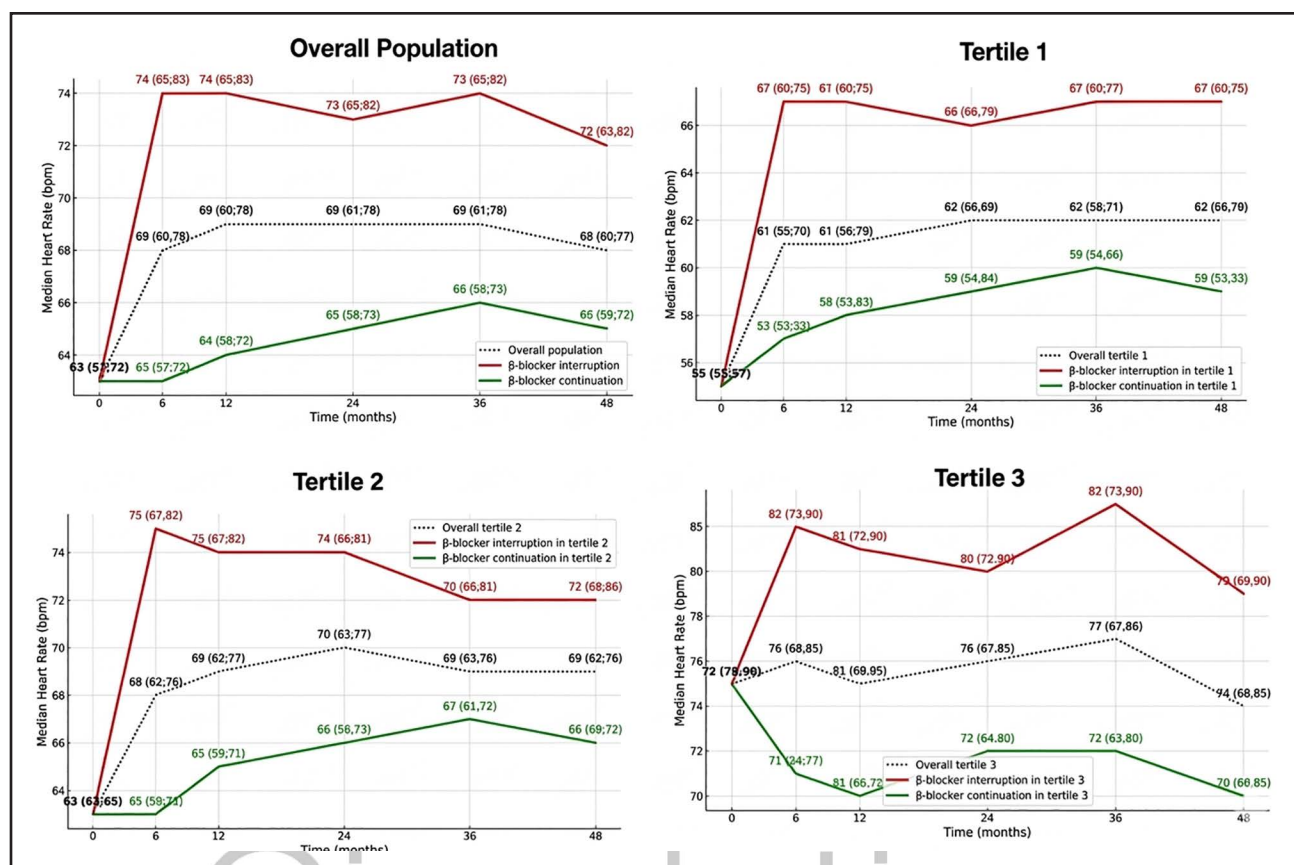


Figure 2. Evolution of median baseline HR according to randomization group.

Red line refers to the interruption group. Green line refers to the continuation group. Dotted line is the overall population or tertile population. HR is expressed as median beats per minute with IQR 25th to 75th. HR indicates heart rate; IQR, interquartile range; and KM, Kaplan-Meier.

and active smokers, and they had risk factors associated with enhanced sympathetic activity such as active smoking or diabetes; (2) post-MI patients with elevated HR had a higher risk of cardiovascular events; (3) there was no interaction between baseline HR and the effect of β -blocker interruption on outcomes, supporting the continuation of therapy regardless of baseline HR; and (4) β -blocker interruption was associated with a significant HR increase, often exceeding 10 bpm, especially in patients previously treated with high doses.

Patients in the highest HR tertile were younger, more frequently female, and had a higher prevalence of smoking and cardiovascular risk factors such as diabetes and renal insufficiency consistent with previous cohort observations.^{15–17} This profile suggests a population with elevated baseline cardiovascular risk and heightened sympathetic activity, leading to nearly a 2-fold increase in hard cardiovascular events in this stable post-MI population. The fact that these patients were prescribed higher β -blocker doses at baseline indicates an attempt to control their HR aggressively; yet, this same strategy may have predisposed them to a greater risk of cardiovascular events upon discontinuation. Abrupt β -blocker discontinuation, particularly in patients with initially high HR, may trigger exaggerated sympathetic activation, increas-

ing the risk of hospitalization for cardiovascular causes, especially coronary events and angina, as observed in the main ABYSS results. However, the delayed divergence of events over time suggests that mechanisms other than an acute sympathetic rebound after β -blocker discontinuation may be involved. Elevated HR may instead reflect a marker of higher underlying cardiovascular risk, potentially related to persistent autonomic imbalance or residual ischemic burden. Combined with the rise in blood pressure we recently reported, these findings further support maintaining well-tolerated β -blocker therapy in post-MI patients.¹⁸

Elevated HR is well recognized as an adverse prognostic factor after MI. In our study, patients with higher baseline HR under β -blocker therapy experienced worse outcomes, particularly increased rates of all-cause death, cardiovascular death, recurrent MI, and HF. Our results are consistent with previous studies showing a linear rise in fatal MI, death, or stroke with each 5-bpm increase in HR.¹⁹ These data highlight the importance of HR control in post-MI patients with an LVEF >40% to prevent adverse cardiovascular events, including HF. Support for the protective effect of β -blockers in post-MI patients with preserved LVEF at risk of HF also comes from recent meta-analyses of the BETAMI, DANBLOCK,

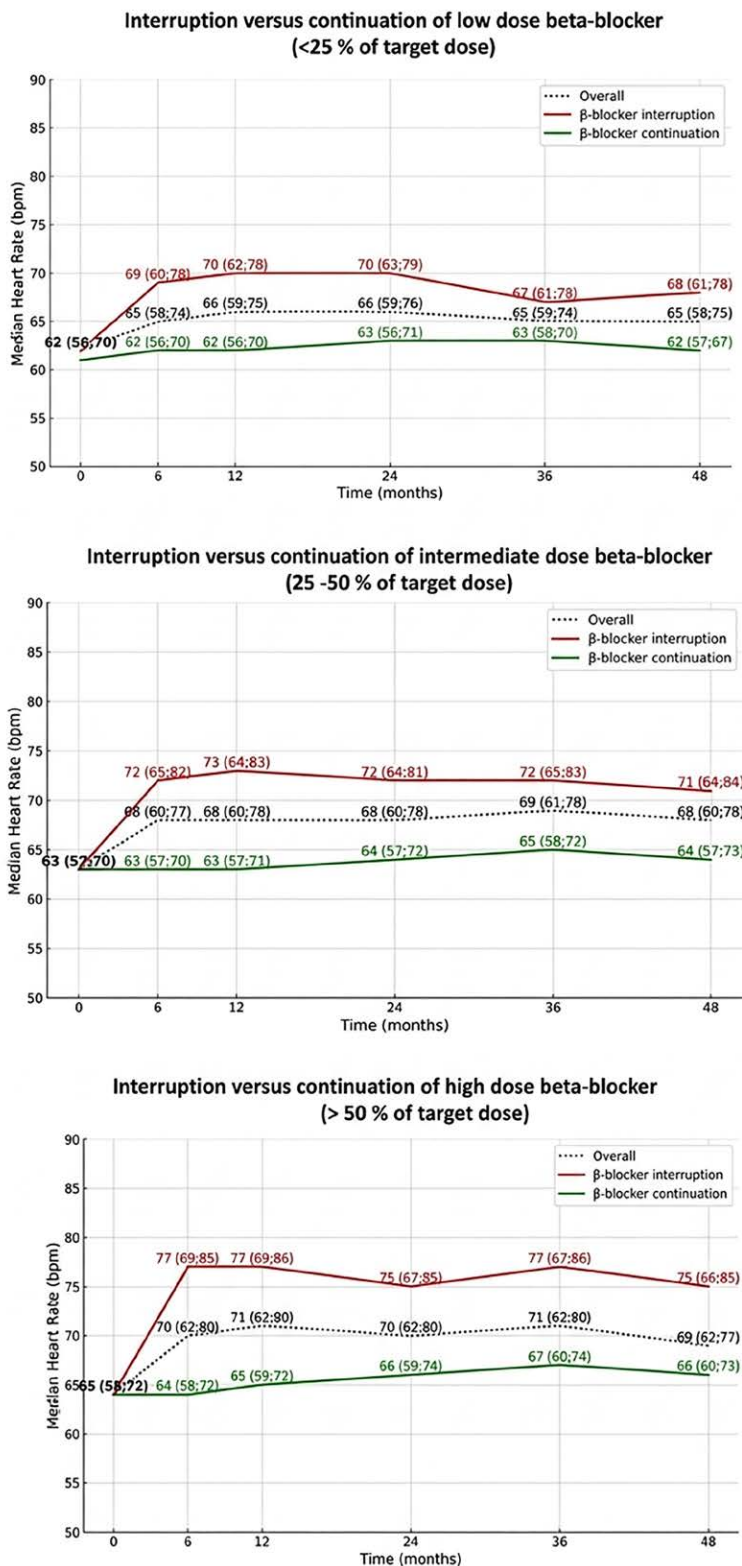


Figure 3. Evolution of median baseline HR according to randomization group and β -blocker dose.

Red line refers to the interruption group. Green line refers to the continuation group. Dotted line is the overall population or tertile population. HR is expressed as median beats per minute with IQR 25th to 75th. HR indicates heart rate; IQR, interquartile range; and KM, Kaplan-Meier.

and CAPITAL-RCT trials.²⁰ Recent studies have emphasized the importance of distinguishing between patients with a normal LVEF ($\geq 50\%$) and a mildly reduced LVEF (40% to 49%) after myocardial infarction. Recent analyses have demonstrated conflicting results regarding

benefit of β -blocker therapy in patients with an LVEF $>50\%$ compared with patients with mildly reduced LVEF remains uncertain for whom β -blockers were associated with a reduction of cardiovascular events. In the present analysis, the effects of β -blocker interruption versus

continuation were consistent across these 2 LVEF categories.^{21,22} These findings therefore support the growing body of evidence questioning the routine long-term use of β -blockers in post-MI patients without overt systolic dysfunction. Interestingly, despite initial concerns regarding QoL, we found no meaningful QoL benefit from β -blocker discontinuation, even in patients with lower HR. Patients with higher baseline HR reported worse QoL scores, suggesting that persistent tachycardia may impair well-being. This challenges the idea that β -blocker interruption improves patient-perceived health and underscores the need for individualized treatment decisions. Slightly lower baseline QoL in high-HR patients also suggest their cardiovascular risk burden contributes to perceived health status beyond HR control.

Limitations

This study was a prespecified secondary analysis of the ABYSS trial. Although the results provide valuable insights, they should be interpreted within the limitations of a secondary analysis. In addition, adherence to β -blocker therapy before randomization was not monitored, which may have led to artificially low HR values in some continuation-arm patients who improved adherence after enrollment. Finally, QoL was assessed using the EQ-5D-3L questionnaire, a generic health status instrument that may not capture more subtle or disease-specific symptoms potentially associated with β -blocker therapy.

Conclusions

In this prespecified secondary analysis of the ABYSS trial, HR remained a critical determinant of major cardiovascular events and mortality in stabilized post-MI patients with LVEF >40%. Elevated HR was associated with a higher risk of cardiovascular adverse events. Abrupt β -blocker discontinuation appeared harmful irrespective of baseline HR and was associated with a significant and sustained HR increase.

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Received November 23, 2025; accepted May 22, 2026.

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Sources of Funding

The ABYSS study was conducted by the ACTION Group (www.action-groupe.org) at Pitié-Salpêtrière Hospital in Paris, France, and funded by a grant from the Programme Hospitalier de Recherche Clinique – PHRC 2015 (French Ministry of Health PHRC 15-0542). The study was sponsored by Assistance Publique – Hôpitaux de Paris.

Disclosures

Guillaume Cayla reports consulting fees from Abbott Vascular, Edwards Lifesciences, MicroPort CRM, and Shockwave Medical, and grants or contracts from Amgen, AstraZeneca, Bayer Healthcare, Biotronik, Medtronic, Pfizer, and Sanofi Pasteur. Grégoire Rangé reports consulting fees from Abbott Vascular. Etienne Puymirat reports consulting fees from Abbott Fund, Amgen, AstraZeneca, Bayer, Biotronik, Boehringer Ingelheim, Bristol-Myers Squibb, Novartis, and Sanofi, and grants or contracts from Abbott Fund. Thomas Cuisset reports consulting fees from Abbott Laboratories, Boston Scientific, Edwards Lifesciences, Medtronic, and Terumo. Eric Vicaud reports consulting fees from Abbott Diabetes Care and Coloplast. Johanne Silvain reports consulting fees from Abbott Vascular, AstraZeneca, Bayer, CSL Behring, and Sanofi-Aventis France. Michel Zeitouni, Niki Procopi, Emile Ferrari, Nicolas Delarche, Paul Guedeney, Olivier Varenne, Romain Cador, Pascal Motreff, Luc-Philippe Christiaens, Anne Bellemain-Appaix, Maxime Fayard, Gilles Bayet, Jean-Michel Quédillac, Pascal Goube, Marc Goralski, Simon Elhadad, Frédéric Hélot, Christophe Caussin, Jean-Charles Aisenfarb, Jean Litalien, Geoffroy Rambaud, David Attias, Raphaëlle Dumaine, Michel S. Slama, Mohamad El Kasty, Laurent Payot, Karim Aacha, Abdourahmane Diallo, and Gilles Montalescot report no conflicts of interest. All authors certified that their disclosure information was complete and accurate.

Supplemental Material

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